AI -POWERED DATA ANONYMIZATION FOR DPEARTMENT OF CENSUS AND STATISTICS OF SRI LANKA

Project ID: 25-26J-123

Project Proposal Report

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NLP FOR IDENTIFIERS AND QUASI-IDENTIFIERS DETECTION IN STRUCTURED CENSUS MICRODATA

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DECLARATION

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We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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…….. ……2025/9/18

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ABSTRACT

The exponential growth of data collected through population censuses raises serious privacy concerns when individual-level microdata is released for research and policy-making. While traditional Statistical Disclosure Control (SDC) techniques address these issues by anonymizing datasets, their effectiveness largely depends on accurately identifying direct identifiers and quasi-identifiers that can lead to re-identification of individuals. Manual and heuristic-based approaches are often error-prone, either overlooking hidden identifiers or overestimating quasi-identifiers, which reduces data utility.

This research focuses on developing a Natural Language Processing (NLP)-driven framework for the detection of identifiers and quasi-identifiers in structured census microdata. The proposed method integrates rule-based detection for explicit identifiers with a transformer-based NLP model that classifies attributes using semantic context, column metadata, and value distributions. It further incorporates contextual risk analysis to validate potential quasi-identifiers, ensuring a balance between privacy preservation and information utility.

The outcome of this work will be a systematic, automated, and context-aware solution that improves upon existing methods by minimizing false positives and false negatives in identifier detection. The feasibility of the proposed approach will be evaluated using real-world census microdata from Sri Lanka, along with synthetic datasets for validation. The framework will directly support anonymization processes such as k-anonymity and l-diversity, thereby contributing to both academic research and practical data governance.

Keywords: Data Anonymization, Statistical Disclosure Control, Quasi-Identifiers, Natural Language Processing, Census Microdata

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LIST OF ABBREVIATIONS

Abbreviation	Word/Description
AI	Artificial Intelligence
ARX	Anonymization Risk Analysis Tool
BERT	Bidirectional Encoder Representations from Transformers
CASSED	Context-based Approach for Structured Sensitive Data Detection
DP	Differential Privacy
GDPR	General Data Protection Regulation
LLM	Large Language Model
ML	Machine Learning
NLP	Natural Language Processing
NIC	National Identity Card
PII	Personally Identifiable Information
QI	Quasi-Identifier
SA	Sensitive Attribute
SDC	Statistical Disclosure Control
SQL	Structured Query Language

Table 1: List of Abbreviation

1. INTRODUCTION

Census microdata refers to individual-level data collected during national surveys and censuses. These datasets provide valuable insights for researchers and policymakers, but they also carry significant privacy risks. To address these risks, it is essential to identify which attributes in the dataset could reveal or contribute to the re-identification of individuals. In data privacy, these attributes are broadly classified into two categories: Identifiers and Quasi-identifiers.

Identifiers are attributes that directly and uniquely reveal the identity of an individual. Examples include the National Identity Card (NIC) number, passport number, full name, phone number, or email address. If such attributes are present in a dataset, they can immediately link a record to a specific individual without the need for additional information. Their detection and removal or masking is therefore a fundamental requirement in any anonymization process.

Quasi-identifiers (QIs) are attributes that do not directly reveal an identity but can contribute to re-identification when combined with other information. For instance, attributes such as age, gender, occupation, district, or marital status may appear harmless in isolation, but when used together, they can significantly narrow down the pool of potential individuals in a dataset. A well-known challenge in Statistical Disclosure Control (SDC) is that quasi-identifiers are context-dependent: an attribute that is safe in one dataset may become risky in another, depending on its distribution and linkage possibilities with external sources.

The detection of identifiers and quasi-identifiers is thus a critical step in the anonymization process. However, existing approaches often rely on manual identification or simple heuristics, which are both time-consuming and prone to inconsistency. In large-scale datasets such as census microdata, this manual process becomes impractical. To overcome these limitations, this component proposes an automated detection framework that integrates both rule-based and Natural Language Processing (NLP)-based techniques. Rule-based methods, such as regular expressions, will be used for detecting explicit identifiers, while NLP models will

analyze column headers, metadata, and value distributions to classify attributes as identifiers, quasi-identifiers, or sensitive fields.

The proposed framework will further incorporate risk validation techniques such as k-anonymity, ℓ-diversity, and uniqueness analysis to confirm whether identified attributes present real disclosure risks. The output of this process will be a risk report and a processed dataset ready for anonymization in established SDC tools such as ARX and sdcMicro. This ensures that privacy protection is carried out systematically, while retaining the analytical value of the census microdata.

1.1 Background

The rapid expansion of data-driven decision-making in government, healthcare, finance, and business has made the availability of microdata increasingly important for research and policy development. Census microdata, in particular, provides valuable insights into demographic, social, and economic patterns that support evidence-based planning. However, releasing such detailed records raises significant concerns about the privacy of individuals, as the data may contain information that can directly or indirectly identify respondents [1].

Data anonymization has emerged as a primary safeguard to ensure that personal data can be shared for secondary use without compromising individual privacy. It refers to the process of transforming datasets so that sensitive information cannot be linked back to specific individuals, while preserving as much analytical utility as possible [2]. Traditional anonymization techniques include suppression, generalization, perturbation, and noise addition, which aim to balance privacy with usability [3].

Within this context, Statistical Disclosure Control (SDC) plays a central role. SDC is a framework of methods applied to microdata to minimize disclosure risk while maximizing data utility [4]. A key requirement in SDC is identifying which attributes are direct identifiers (e.g., names, ID numbers) and which are quasi-identifiers (e.g., age, gender, location). Direct identifiers explicitly reveal identity, whereas quasi-identifiers do not individually identify a person but can lead to re-identification when

combined with external information [5]. The well-known example by Sweeney demonstrated that 87% of the U.S. population could be uniquely identified by the combination of date of birth, gender, and ZIP code [6], underscoring the risks of quasi-identifiers.

Despite advances in anonymization techniques, the correct identification of direct and quasi-identifiers remains a challenging task. Traditional approaches rely heavily on expert judgment or heuristic risk measures, which are often inconsistent, labor-intensive, and prone to either underestimation or overestimation of disclosure risk [7]. Inaccurate identification can lead to two critical problems: (i) privacy breaches, if risky attributes are overlooked, and (ii) loss of data utility, if too many attributes are unnecessarily anonymized [8].

Recent research highlights the potential of Natural Language Processing (NLP) and machine learning (ML) techniques to address this challenge. These approaches can automatically analyze attribute names, metadata, and value patterns to classify fields as identifiers, quasi-identifiers, or sensitive attributes [9]. Applying such techniques to census microdata provides an opportunity to develop systematic, context-aware solutions that strengthen privacy protection while maintaining the analytical richness of released datasets.

1.2 Literature Survey

Foundations: SDC and privacy models.

Statistical Disclosure Control (SDC) provides the methodological basis for releasing microdata with minimized disclosure risk while preserving analytical utility, building on classical disclosure risk concepts and practices in official statistics [1], [4]. Canonical privacy models—k-anonymity, ℓ-diversity, t-closeness, and differential privacy—shape how protection is specified and audited: k-anonymity constrains re-identification by equivalence classes; ℓ-diversity and t-closeness target attribute disclosure within classes; and differential privacy offers distribution-level protection via calibrated noise [6], [11], [12], [10]. All rely on a correct delineation of direct identifiers and quasi-identifiers (QIs) that may enable linkage with external sources [1], [5], [6].

Heuristic and risk-based identification of QIs.

Common practice operationalizes QI identification via expert rules and risk heuristics—per-attribute uniqueness, special uniques detection (SUDA2), and k-map–style uniqueness checks on candidate combinations—providing pragmatic cues for linkage risk assessment in microdata releases [4], [18]. Recent work augments this with automated risk scoring/selection, estimating per-attribute re-identification probabilities and proposing QI sets that reduce linkage risk after anonymization while controlling information loss [7], [8].

Rule-based and DLP pipelines for direct PII.

Direct identifiers (e.g., name, phone, email, national ID) are effectively captured by deterministic recognizers (regex, checksum validation) and NER components in data-loss-prevention (DLP) pipelines. Toolkits such as Microsoft Presidio combine pattern recognizers with NLP entity models and support domain-specific custom recognizers—useful for national formats and local address conventions—yielding high precision for overt PII in both free text and structured columns [17]. These pipelines complement, rather than replace, QI identification.

ML for semantic typing of structured columns.

To move beyond hand-crafted patterns, semantic data type detection treats each column as a unit to be classified from header text, character/value distributions, and token patterns. Deep models such as Sherlock showed that learned representations can robustly classify column semantics (e.g., person names, cities, ZIPs), providing a data-driven basis to flag likely personal/sensitive fields across heterogeneous tables [16].

Context-aware NLP for sensitive data in tables.

Recent approaches cast column classification as an NLP problem over headers and sampled values, allowing transformers to exploit intra-column semantics. CASSED frames structured sensitive-data detection as multi-label column classification with transformer encoders, improving generalization over noisy headers and coded values relative to rule-only baselines [9]. In parallel, table-aware pretraining (e.g., TaBERT, TABBIE) equips models to reason jointly over textual and tabular signals, strengthening detection of columns whose semantics emerge from value patterns and schema context [19], [20].

Automated QI recognition with SDC integration.

Algorithmic QI recognition explicitly links detection to SDC practice: graph-/risk-based QI recognition derives the QI set and its dimensionality to balance identity-disclosure reduction with utility retention [7]; automated QI selection scores attributes by attacker success probabilities and validates reductions in post-anonymization linkage risk on benchmark data [8]. In practical pipelines, these detectors integrate with SDC tools such as ARX and sdcMicro, which provide k-anonymity/ℓ-diversity/t-closeness optimization, risk analyzers (sample-uniques, prosecutor/journalist/marketer models), and utility metrics for iterative tuning [14], [15].

1.3 Research Gap

In reviewing the past years of research on identifier and quasi-identifier detection in statistical disclosure control (SDC), several limitations are evident. Classical approaches such as manual expert-driven QI selection [1], [4], [5], and heuristic uniqueness-based metrics [6], [18] provide only partial solutions. They often result in either under-identification of risky attributes (leading to disclosure risk) or over-identification (causing unnecessary data loss).

Machine learning and semantic column typing approaches [16], [9] improve automation but still face challenges in multi-column context integration and local adaptability. For example, transformer-based models such as CASSED [9] are effective for sensitive data detection but primarily focus on single-column classification, overlooking cross-attribute relationships that create re-identification risks. Similarly, while automated QI recognition algorithms [7], [8] provide structured risk-based selection, they rarely incorporate semantic understanding of attribute meaning, and their reliance on global heuristics can misrepresent risks in localized census contexts.

Furthermore, widely used PII detection toolkits like Microsoft Presidio [17] and SDC systems like ARX and sdcMicro [14], [15] lack built-in intelligence for context-aware QI detection; they rely on user-specified identifiers. This highlights a clear gap: there is no unified, context-aware framework that combines rule-based detection for direct PII, NLP-driven classification for quasi-identifiers, and risk validation integrated with SDC tools.

Key Requirement / Feature	**Manual/Heuristic Approaches [1], [4], [5], [6], [18]**	**Automated Risk Scoring [7], [8]**	**Semantic Typing & ML (Sherlock, CASSED) [16], [9]**	**Existing Tools (Presidio, ARX, sdcMicro) [14], [15], [17]**	**Proposed System**
Reliance on expert knowledge vs. automation	✗ Mostly manual	✓ Partial automation	✓ Automated classification	✗ User-dependent	✓ Fully automated hybrid
Identifies Direct Identifiers (PII) (names, IDs, emails, etc.)	✓ Simple/obvious	✗ Focus on QIs only	✗ Limited	✓ Pattern-based	✓ Rule-based + NLP
Identifies Quasi-Identifiers (QIs) (context-dependent attributes)	✓ Heuristic uniqueness	✓ Risk scoring	✓ Semantic - based	✗ Relies on manual specification	✓ NLP-driven + context validation
Handles multi-column combinations (e.g., Age + District + Occupation)	✗ Limited	✗ Limited	✗ Mostly per-column	✗ Not automated	✓ Context-aware integration
Incorporates external/public data linkability	✗ No	✗ No	✗ Rare	✗ No	✓ Extendable with external checks
Local adaptability (e.g., Sri Lanka census formats, NIC numbers, local names)	✗ Weak	✗ Generic	✗ Generic	✗ Requires manual regex	✓ Customizable recognizers + NLP embeddings
Integration with full SDC pipeline (risk analysis + anonymization techniques)	✗ Weak link	✓ Partial	✗ Standalone classification	✓ Integrated (but without detection intelligence)	✓ Unified detection + SDC integration

Table 2: Research Gap of the Component

1.4 Research Problem

Despite significant progress in data anonymization and Statistical Disclosure Control (SDC), the accurate identification of direct identifiers and quasi-identifiers in census microdata remains an unresolved challenge. Existing approaches are mini sub problems like below,

- **Rely heavily on manual selection and heuristics** [1], [4], [6], which are inconsistent, time-consuming, and prone to overlooking subtle quasi-identifiers or overestimating risk.
- **Focus on isolated attribute analysis**, neglecting multi-column interactions and contextual relationships that can enable re-identification [5], [9], [16].
- **Lack adaptability to local datasets**, such as Sri Lankan census microdata, where domain-specific identifiers (e.g., NIC numbers, locality names) are not well recognized by global models and tools [15], [17].
- **Provide limited integration with SDC pipelines**, leaving a gap between identifier detection and practical anonymization implementation [7], [8], [14].

As a result, current methods either fail to fully protect privacy (by missing risky attributes) or unnecessarily reduce data utility (by over-generalizing safe attributes). This dual risk creates a critical barrier for organizations seeking to release high-quality census microdata for research while ensuring compliance with privacy regulations.

Therefore, the research problem is:

How can we design a systematic, automated, and context-aware framework that accurately detects direct identifiers and quasi-identifiers in structured census microdata, while integrating seamlessly with SDC techniques to balance privacy protection and data utility?

2. OBJECTIVES

2.1 Main Objective

The main objective of this research is:

To develop a systematic and automated Natural Language Processing (NLP)–based framework for the accurate detection of direct identifiers and quasi-identifiers in structured census microdata, in order to support Statistical Disclosure Control (SDC) processes that ensure individual privacy while preserving maximum data utility.

2.2 Specific Objectives

The specific objectives of this research are:

To conduct a comprehensive review of existing methods for identifier, quasi-identifier, and sensitive attribute detection.
This involves critically examining literature over the past 15 years, covering traditional heuristic methods, statistical disclosure control techniques, and recent NLP/ML-driven approaches. The review will highlight the capabilities, limitations, and applicability of these methods to structured census microdata. The output will provide a clear foundation and justification for developing an improved framework.

To design a hybrid framework that integrates rule-based PII detection with NLP-based quasi-identifier classification.
The framework will combine deterministic recognizers (e.g., regex, dictionaries, named entity recognition) for direct identifiers with transformer-based NLP models for quasi-identifiers and sensitive attributes. The design will include how metadata, value distributions, and semantic context will be used, ensuring both accuracy and adaptability to different census data structures.

To implement the proposed framework using suitable data science and NLP tools.
This objective will focus on the practical development of the system using Python libraries such as HuggingFace Transformers, spaCy, and Presidio, together with statistical disclosure control tools like ARX and sdcMicro. The system will be prototyped on real Sri Lankan census microdata (in English) and, where necessary, supplemented with synthetic datasets for model training and validation.

To evaluate the framework's effectiveness in terms of accuracy, scalability, and balance between privacy and data utility.
The framework will be assessed using well-defined metrics such as precision/recall for identifier detection, k-anonymity/ℓ-diversity risk scores, and information loss measures. Comparative analysis will be carried out against existing heuristic and ML-based methods to demonstrate improvements. The evaluation will also test

scalability across datasets of different sizes to ensure the framework's feasibility in real-world census settings.

To assess the feasibility of integrating the framework with established SDC techniques and anonymization pipelines.
Beyond identifier detection, this objective ensures that the output of the proposed system can be directly applied in anonymization processes. The feasibility study will include practical tests with anonymization techniques such as generalization, suppression, and differential privacy, demonstrating that the framework provides actionable inputs to protect privacy while maintaining dataset usability.

To document findings, derive insights, and propose future improvements.
Finally, the research will systematically document the methodology, experiments, and outcomes, ensuring transparency and reproducibility. Recommendations will be provided for extending the framework to multilingual datasets, scaling to national census systems, and integrating with broader data governance practices. This ensures that the project contributes not only a working prototype but also long-term value to the field of statistical disclosure control.

3. METHODOLOGY

The proposed research follows a hybrid methodology that combines rule-based detection for direct identifiers and NLP-driven classification for quasi-identifiers in structured census microdata. The system will operate as a modular pipeline with three main stages:

1. **Data Input and Preprocessing** – Census microdata (English language) will be cleaned, standardized, and formatted for analysis.
2. **Identifier Detection Layer** – Direct identifiers will be captured through deterministic recognizers (e.g., regex for emails, NIC patterns, phone numbers) while candidate quasi-identifiers will be extracted based on metadata, attribute headers, and value distributions.
3. **NLP Classification and Risk Validation** – Transformer-based models will classify attributes as identifiers, quasi-identifiers, or sensitive attributes. Detected attributes will be validated using Statistical Disclosure Control (SDC) risk measures (e.g., k-anonymity, l-diversity), ensuring integration with anonymization tools.

3.1 Overall System Diagram

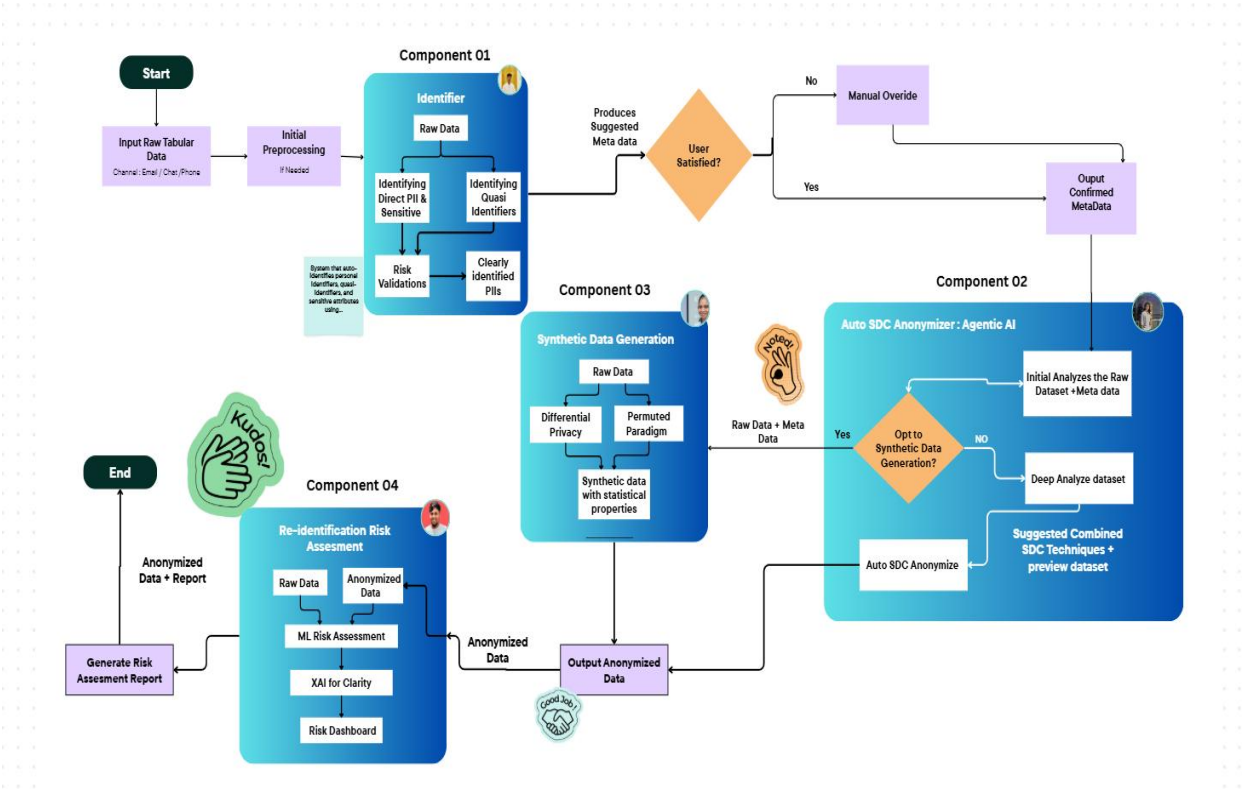

Figure 1:Overall System Diagram

Workflow for Data Anonymization

This framework explains how census microdata is processed, anonymized, and evaluated by the different members of the project team. Each component is handled by a dedicated member, ensuring that identifiers and quasi-identifiers are systematically detected, anonymization is applied, synthetic data is generated when necessary, and risks are continuously validated before final data release.

Member One – Identifier Detection

- The raw census microdata is first examined to detect **direct identifiers** (e.g., name, NIC, phone number), **quasi-identifiers** (e.g., age, district, occupation), and **sensitive attributes** (e.g., salary, health conditions).

- Rule-based methods (e.g., regex, Microsoft Presidio) are combined with **NLP models** (Transformers + spaCy) to recognize attributes more accurately, especially when quasi-identifiers are context-dependent.
- Detected attributes are validated through basic **risk checks** (uniqueness percentages, frequency distributions) to confirm whether they pose re-identification risks.

Member Two – Anonymization Techniques

- The findings from Member One are analyzed to determine the most suitable **Statistical Disclosure Control (SDC)** techniques.
- Depending on the dataset, methods such as **generalization, suppression, masking, or aggregation** are applied.
- An **AI-assisted anonymizer** suggests combinations of techniques while balancing **privacy protection and data utility**.
- If anonymization alone is insufficient or heavily distorts the dataset, the process is handed over to Member Three for **synthetic data generation**.

Member Three – Synthetic Data Generation

- When recommended, **synthetic data** is generated to replace risky attributes or entire tables while preserving statistical properties.
- Approaches such as **Differential Privacy** and **permuted paradigm methods** are applied to ensure realistic but privacy-preserving data.
- The synthetic dataset is then merged with safe attributes from the original data to produce a **privacy-protected version** suitable for research.

Member Four – Risk Evaluation & Validation

- The anonymized or synthetic dataset undergoes **re-identification risk assessment** using ML-based techniques.
- A **risk dashboard** is generated to show the likelihood of re-identification and the effectiveness of applied techniques.

- If risk is above the threshold, feedback is sent back to Members One and Two for further adjustments.
- If acceptable, the dataset is finalized and a **Risk Assessment Report** is produced for documentation and compliance.

Outcome: This structured workflow ensures that census microdata can be anonymized while maintaining research usefulness. Direct identifiers, quasi-identifiers, and sensitive attributes are systematically handled, with risk validation acting as the final safeguard before release.

3.2 Research Tasks and Workflow

The research will be conducted in the following phases:

1. **Literature Review and Problem Framing** – A thorough review of methods for identifier and QI detection, along with current SDC practices, to refine the scope and identify baseline models/tools.
2. **Framework Design** – Conceptualizing the hybrid model that integrates rule-based recognizers with NLP models.
3. **Prototype Development** – Implementing the detection framework using Python, HuggingFace Transformers, spaCy, Microsoft Presidio, and anonymization tools like ARX/sdcMicro.
4. **Evaluation Phase** – Assessing model performance using accuracy, precision, recall, F1-score, as well as risk/utility metrics such as k-anonymity and information loss.
5. **Feasibility and Integration Testing** – Ensuring compatibility of the detection system with standard SDC pipelines, focusing on practicality and efficiency.
6. **Documentation and Analysis** – Writing up results, conducting error analysis, and framing recommendations for future work.

3.3 Data Collection

The primary dataset will be census microdata obtained from the Department of Census and Statistics, Sri Lanka (in English). For testing adaptability, synthetic data may also be generated to simulate additional identifiers and quasi-identifiers. All datasets will be anonymized and ethically managed, adhering to data protection guidelines.

3.4 Tools and Technologies

- **Programming & ML:** Python, HuggingFace Transformers, spaCy.
- **Pattern-based Detection:** Microsoft Presidio, regex libraries.
- **SDC Tools:** ARX (Java), sdcMicro (R).
- **Data Handling:** Pandas, NumPy.
- **Evaluation:** Scikit-learn, statistical risk measures.
- **Project Management:** WBS and Gantt chart for tracking progress.

3.5 Expected Outcomes of Methodology

- A functioning prototype capable of detecting direct identifiers and quasi-identifiers in census microdata automatically.
- Quantitative evaluation of privacy protection vs. data utility.
- Demonstration of integration with SDC anonymization pipelines.
- Feasibility analysis showing real-world applicability in the Sri Lankan census context.

3.6 Individual Diagram

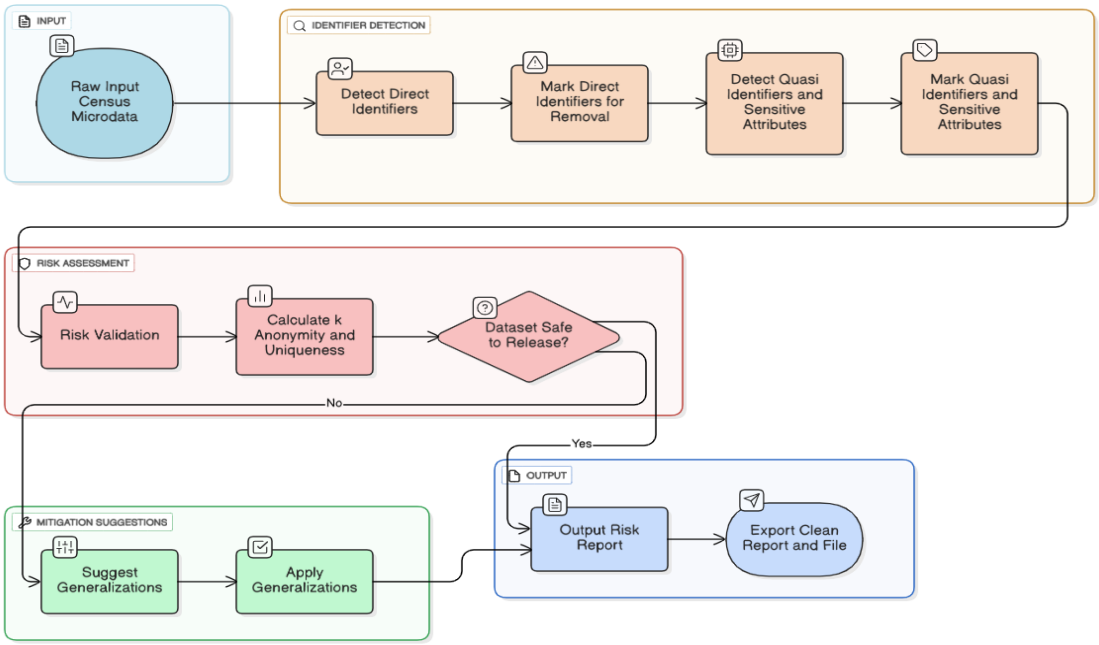

Figure 2: Individual System Diagram

Explanation of Workflow

1. Data Input

The system begins by receiving structured census microdata, typically provided in CSV or Excel formats. At this stage, a secure interface ensures that the dataset is uploaded safely. This input forms the foundation of the entire anonymization process, as the quality and structure of raw data directly affect the effectiveness of subsequent steps.

2. Pre-Processing

Before analysis, the dataset is cleaned and standardized. Pre-processing tasks include handling missing values, normalizing inconsistent column headers, and unifying data formats to maintain consistency. This step ensures that the input is reliable, properly formatted, and ready for identifier detection.

3. Identifier Detection

In this stage, two categories of attributes are detected:

- Direct Identifiers (e.g., names, NIC numbers, phone numbers) are identified using rule-based recognizers and regular expressions. These attributes are explicitly linked to individuals and must be removed or masked immediately.
- Quasi-Identifiers (e.g., age, location, marital status) are detected using Natural Language Processing (NLP)-based classification models. These attributes may not directly reveal identity but can, in combination with other data, increase re-identification risk.

4. Risk Validation

Once identifiers are detected, the attributes are tested against statistical disclosure control measures. Metrics such as k-anonymity, uniqueness, and re-identification risk are calculated to determine how easily individuals could be traced back. This step provides a quantitative validation of the dataset's safety level before release.

5. Integration with SDC Tools

The validated output is connected to established anonymization tools such as ARX or sdcMicro, which apply generalization, suppression, or other SDC techniques to mitigate risks. This integration ensures that the framework is not only automated but also compatible with widely used anonymization standards.

6. Reporting & Output

The final stage produces a comprehensive risk report. Results are visualized in a dashboard that highlights detected identifiers, risky attributes, levels of disclosure risk, and recommended anonymization strategies. The system then exports both the cleaned dataset and the risk report, ensuring that researchers receive data that is anonymized yet still useful for analysis.

3.7 Requirements

3.7.1 Functional Requirements

These requirements specify **what the proposed system should do**:

1. **Data Upload & Preprocessing**
    - The system must allow census officers or researchers to securely upload structured census microdata (CSV/Excel).
    - The system must preprocess datasets by cleaning, normalizing column headers, and handling missing values.
2. **Direct Identifier Detection**
    - The system must automatically detect direct identifiers (e.g., name, NIC, phone number, email, address) using rule-based recognizers and pattern matching.
3. **Quasi-Identifier Detection**
    - The system must classify potential quasi-identifiers (e.g., age, gender, district, occupation) using NLP-based models trained on metadata and value distributions.
4. **Sensitive Attribute Identification**
    - The system must identify attributes that may contain sensitive categories (e.g., religion, ethnicity, income level).
5. **Risk Validation & Integration with SDC**
    - The system must validate detected attributes against disclosure risk measures (e.g., k-anonymity, ℓ-diversity).
    - The output must be compatible with anonymization tools such as ARX and sdcMicro for further anonymization.
6. **Reporting & Visualization**
    - The system must generate reports summarizing detected identifiers and quasi-identifiers.
    - The user must be able to view results via a dashboard, showing risk levels and anonymization readiness.

3.7.2 Non-Functional Requirements

These requirements specify **how the system should perform** rather than what it does:

1. **Performance**
    - The system should process medium-sized census datasets (up to 100,000 records) within reasonable time (< 5 minutes).
    - Identifier detection must achieve high accuracy, with a target precision and recall above 85%.
2. **Scalability**
    - The framework should be designed to handle larger datasets (millions of records) with minimal modifications.
3. **Usability**
    - The front-end dashboard must be simple, intuitive, and usable by non-technical census officers.
    - Results must be clearly explained with risk scores and suggested anonymization actions.
4. **Security & Privacy**
    - Uploaded datasets must be processed securely, ensuring no unauthorized access or leakage.
    - The system must not store personal data permanently; temporary storage should be cleared after analysis.
5. **Adaptability**
    - The system should allow custom recognizers for country-specific identifiers (e.g., Sri Lankan NIC).
    - The NLP model should be adaptable for different census formats.
6. **Reliability**
    - The system must consistently detect identifiers and quasi-identifiers with minimal false positives/negatives.
    - The system must handle incomplete or noisy datasets without failure.

4. Work Break Down Structure

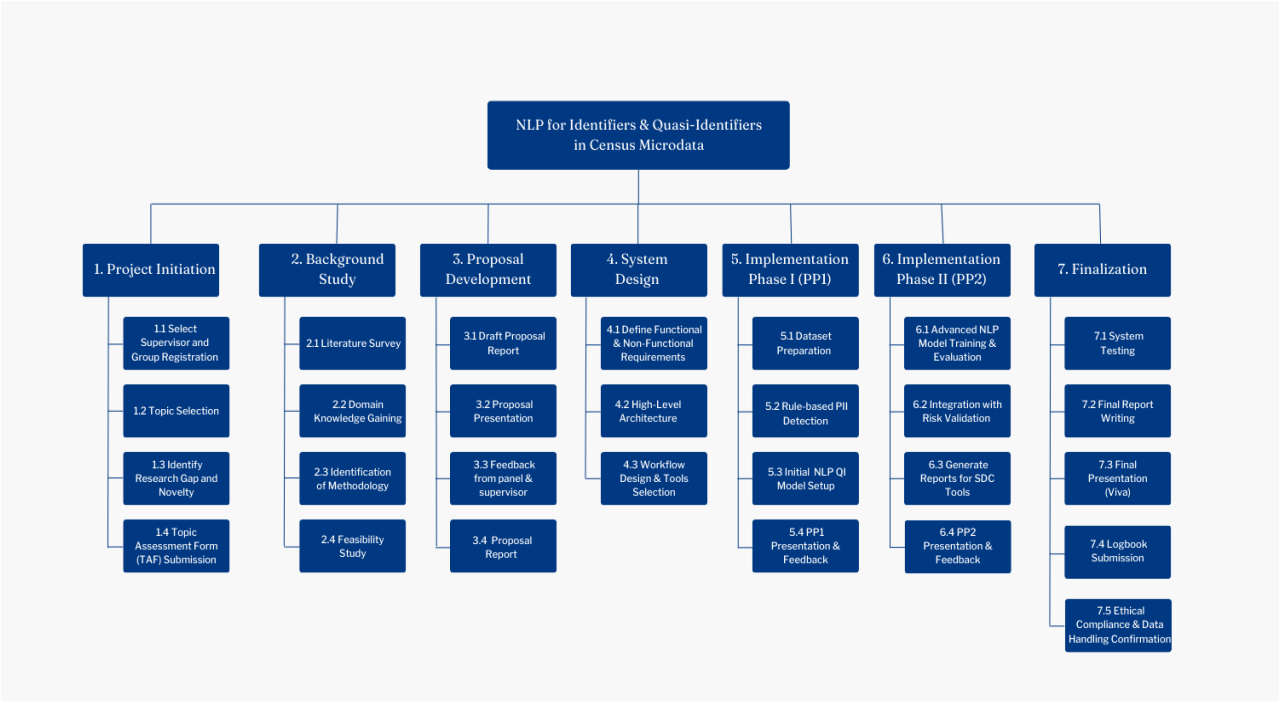

Figure 3: Work Break Down Structure

5. Gantt chart

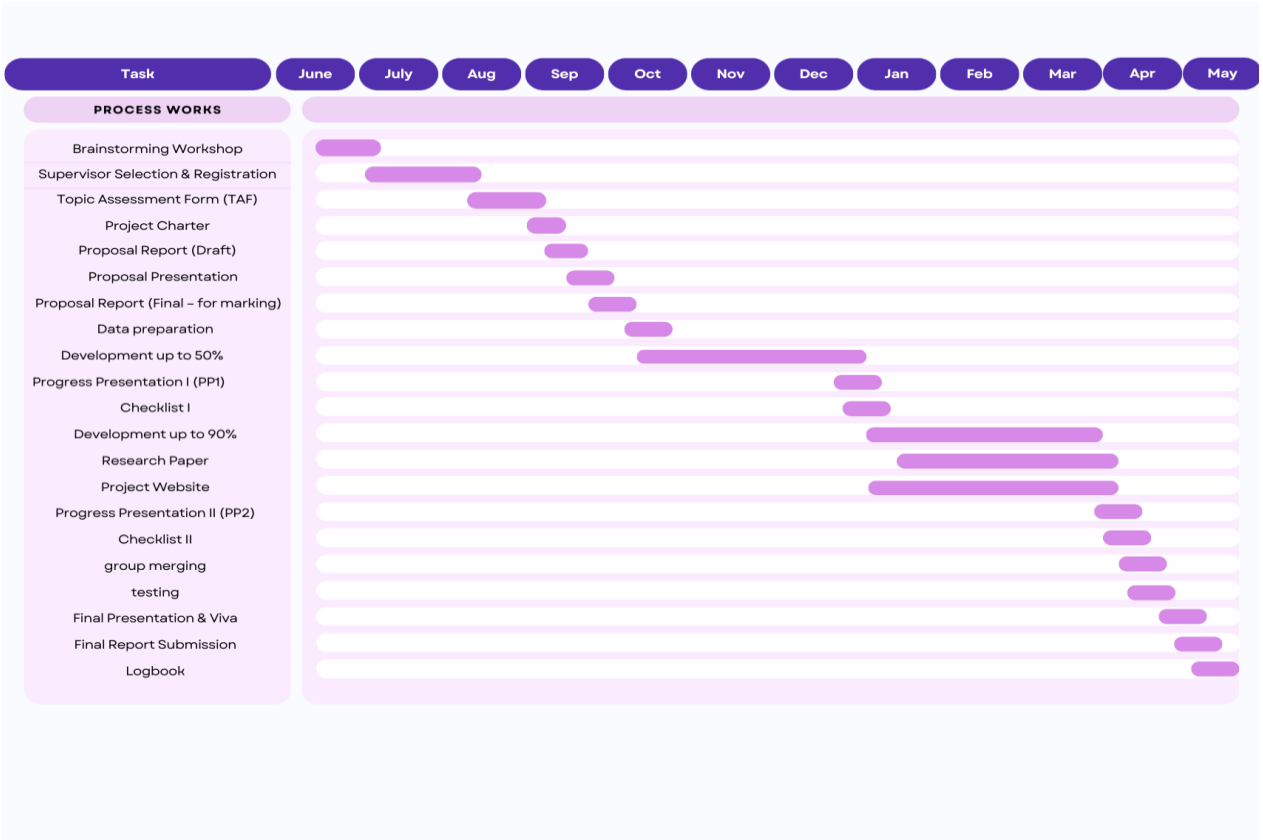

Figure 4: Gantt Chart

6. DESCRIPTION OF PERSONNEL AND FACILITIES

This project is carried out by a group of undergraduate students specializing in Data Science, under the supervision of academic staff from the Faculty of Computing and Information Systems.

For my component, I, I.M.M. Mujahid (Reg. No: [IT22266750]), am responsible for developing the NLP-based Identifier and Quasi-Identifier Detection module. My role primarily involves dataset preparation, preprocessing, application of rule-based methods for direct identifiers, implementation of transformer-based NLP models for quasi-identifier detection, and integration of the detection system with Statistical Disclosure Control (SDC) pipelines. This task requires expertise in natural language processing, machine learning, and software engineering for modular system design, development, and evaluation.

Facilities Available

- **Computational Resources**:
  Access to high-performance computing labs with GPU-enabled workstations that support deep learning model training and experimentation. Cloud-based resources (Google Colab, AWS, or Azure) may also be utilized for scalable testing.
- **Software Tools**:
  Python ecosystem with libraries such as HuggingFace Transformers, spaCy, Scikit-learn, Pandas, NumPy, and Matplotlib for model development and evaluation. Microsoft Presidio for rule-based PII detection and ARX / sdcMicro for anonymization experiments.
- **Data Resources**:
  Access to census microdata provided by the Department of Census and Statistics, Sri Lanka, supplemented with synthetic datasets for testing and validation. Pre-trained NLP models and open-source benchmark datasets will also be used for fine-tuning and comparative evaluation.
- **Collaboration Platforms**:
  GitHub for version control and collaborative development, Google Drive/Cloud storage for managing datasets and project files, and communication platforms such as Slack/MS Teams for coordination within the project group and with supervisors.

7. BUDGET AND BUDGET JUSTIFICATION

DRAFT BUDGET ALLOCATION

Category	Item	Quantity	Unit Cost (Rs.)	Total (Rs.)
Hardware & Infrastructure	Laptop / Workstation (Intel i7, 16GB RAM, 1TB SSD)	1	Already Have	0
Subtotal				**0**
Software & Tools	Python & R Libraries (open-source)	-	Free	0
	Development Tools (VS Code, Jupyter, GitHub)	-	Free	0
	Docker & Kubernetes (open-source)	-	Free	0
Subtotal				**0**
Resources	Research Assistant LLMs	1	6,000	54,000
	Domain Expert Journals that need be purchased	Approx 4	1000	4,000
	Internet Browsing	16GB data per month	1145	9160
Subtotal				**67,160**
Travel	Travel to Department of Census & Statistics (for data access & validation meetings)	10 trips	Rs. 200	2,000
	Travel to SLIIT for Supervisor meetings and physical Team Meetings.	8 trips per month	Rs.200	14,400
Subtotal				**16,400**
Dissemination & Documentation	Printing & Binding of Thesis / Reports		15,000	15,000
	Journal / Conference Publication Fees	1	12,000	12,000
	Miscellaneous (stationery, contingencies)		3000	3000
Subtotal				**30,000**

Table 3: Budget Allocation

Total Estimated Budget: Rs. 113,560

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